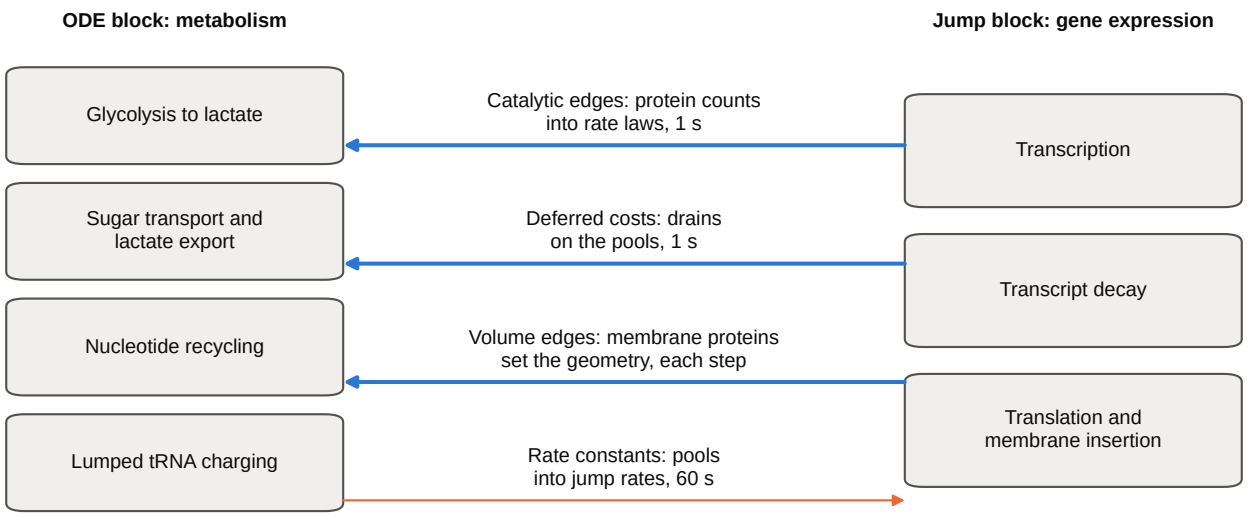
ODE block: metabolism
Jump block: gene expression
Glycolysis to lactate
Sugar transport and lactate export
Nucleotide recycling
Lumped tRNA charging
Catalytic edges: protein counts into rate laws, 1 s
Deferred costs: drains on the pools, 1 s
Volume edges: membrane proteins set the geometry, each step
Rate constants: pools into jump rates, 60 s
Transcription
Transcript decay
Translation and membrane insertion